

Experiment-2

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Slot-B

Title

Implementation of Markov Decision Process for Reinforcement Learning

Aim

To understand and implement a Markov Decision Process (MDP) by defining its components—states, actions, transition probabilities, rewards, and discount factor—and to demonstrate how an agent interacts with an environment.

Theory

A Markov Decision Process (MDP) is a mathematical framework used to model sequential decision-making problems in Reinforcement Learning.

An MDP is represented as:

where:

- $S \rightarrow$ Set of states
- $A \rightarrow$ Set of actions
- $P \rightarrow$ Transition probability
- $R \rightarrow$ Reward function
- γ (Gamma) $\rightarrow$ Discount factor

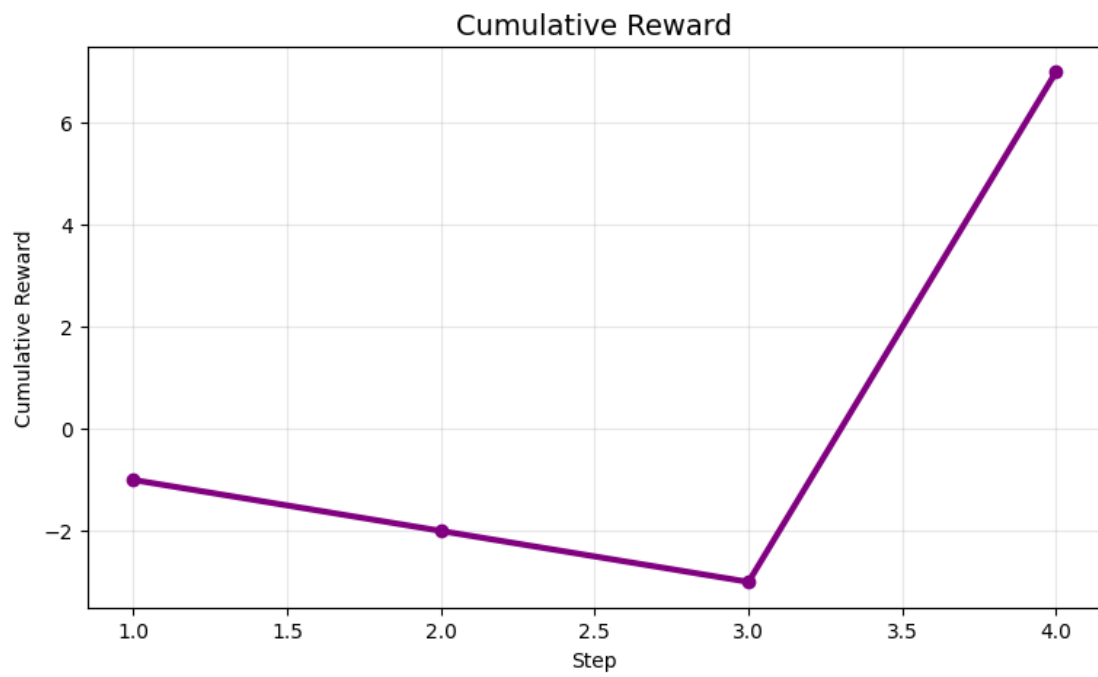
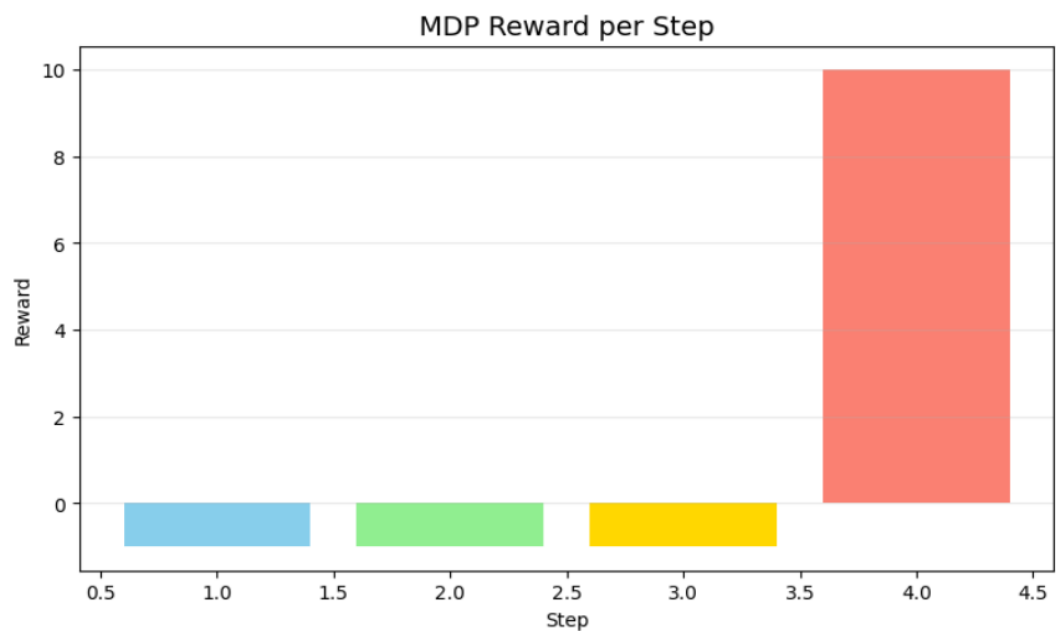
The agent observes the current state, selects an action, receives a reward, and moves to the next state.

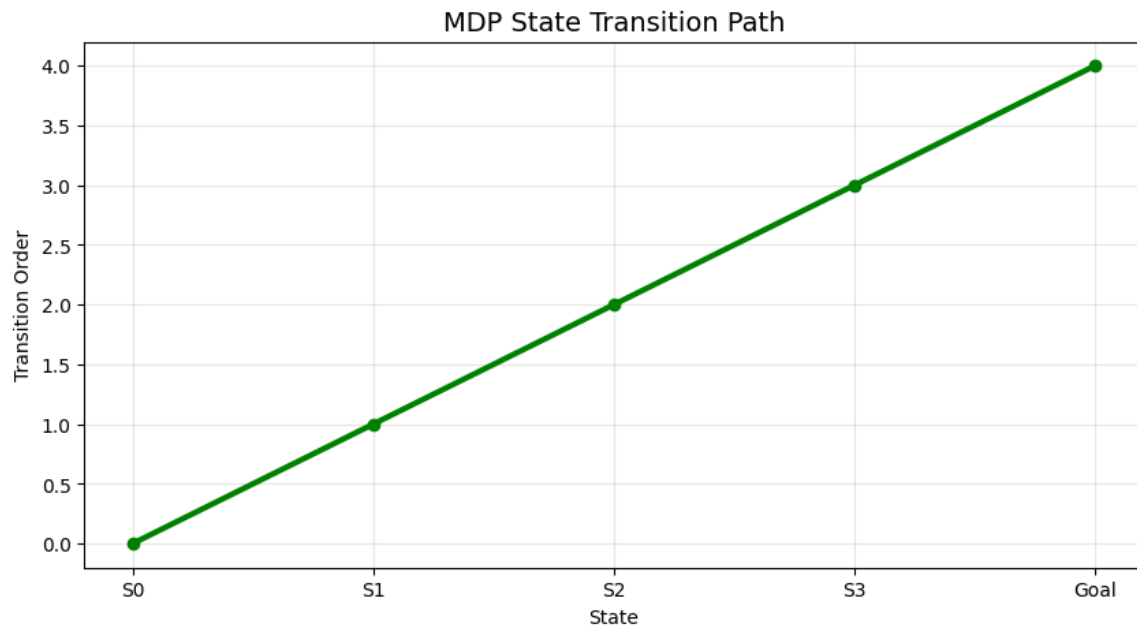
The Markov property means that the next state depends only on the current state and current action, not on the complete history.

Procedure

1. Define the Reinforcement Learning environment.
2. Identify the set of states.
3. Define the possible actions available to the agent.
4. Define the transition probabilities between states.
5. Define the reward function for each state-action transition.
6. Select an appropriate discount factor (γ).
7. Initialize the MDP.
8. Start the agent from an initial state.
9. Select an action from the available action set.
10. Apply the action to the environment.
11. Observe the next state and corresponding reward.
12. Repeat the process until the terminal state is reached.
13. Display the state transitions, actions, and rewards.
14. Calculate the total/discounted reward obtained by the agent.
15. Analyze the MDP and verify the Markov decision-making process.

VISUALIZATIONS





Component	Example
States	S0, S1, S2, S3, Goal
Actions	Up, Down, Left, Right
Transition	Current state → Next state
Reward	+10 for reaching Goal, -1 for an invalid/move action
Discount Factor	$\gamma = 0.9$
Initial State	S0
Terminal State	Goal

Current State	Action	Next State	Reward
S0	Right	S1	-1
S1	Right	S2	-1
S2	Right	S3	-1
S3	Right	Goal	+10

Result

The MDP is successfully defined with states, actions, transition probabilities, rewards, and a discount factor. The agent successfully interacts with the environment, performs state transitions, receives rewards, and reaches the terminal state.